

Research Proposal: AI-Enhanced Elevator Improvement

ML, AI, UI/UX, and HCI Perspectives

1. Working Title

Human-Adaptive Smart Elevator System: Improving Efficiency, Queue Experience, Accessibility, and Safety Using ML, AI, UI/UX, and HCI

2. Abstract

Modern elevators are human-facing intelligent mobility systems. Traditional control uses simple rules (hall calls, direction, position) and ignores adaptive human behavior (e.g., pressing both Up/Down during peaks), causing wasted trips, energy, and poor UX.

We propose a hybrid system combining rule-based control (RDP/WFST), computer vision, time-series learning, and a safety-aware AI gate. The goals are to reduce waiting time, avoid wasteful motion, estimate real queue demand, improve accessibility, and enforce safety-aware dispatch decisions.

3. Problem Statement

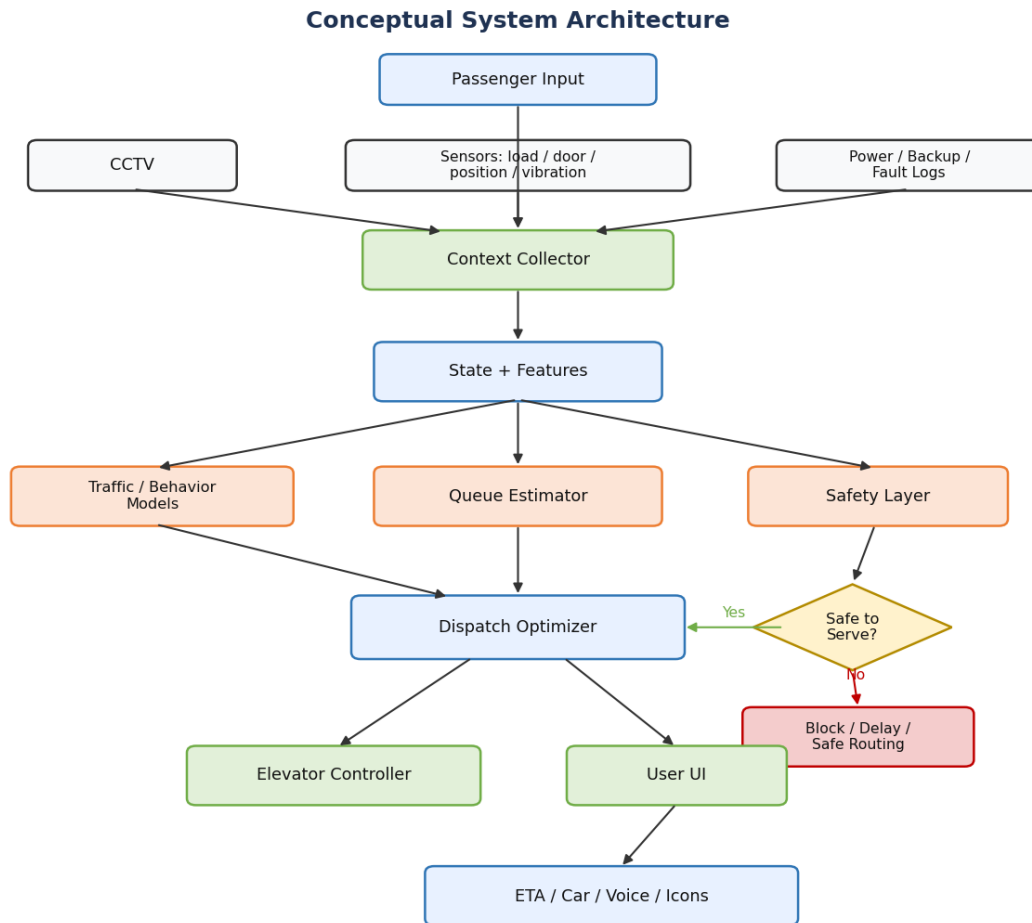
- Human-adaptive behavior vs. rigid machines → button misuse causes wasteful dispatch.
- Invisible demand → button press $\neq$ number of waiting people.
- Partially inclusive HCI → gaps for non-English, low-literacy, and accessibility needs.
- Fragmented safety signals → no unified, real-time decision layer.

4. Research Objectives

- Efficiency Learning — reduce unnecessary movement via hybrid control.
- Smart Queue Dispatch — use CV + temporal models to estimate real demand.
- Inclusive HCI/UI/UX — improve accessibility beyond numbers/text.
- Safety-Aware AI Layer — gate dispatch based on risk.
- Transparent Feedback — ETA, assignment, and explanations.

5. Conceptual System Architecture

The diagram below follows the original Mermaid-style top-down structure. It is inserted as an image so it will print reliably in Word.



6. Refined Work Packages (Modeling Strategy)

WP1 — Efficiency: Hybrid RDP/WFST + Learning

Core Idea: Use deterministic rule-based control with learned cost optimization.

- States: idle, up, down, full, partial
- Inputs: call, capacity, direction
- Transition = cost(time, energy, congestion)
- Example: If elevator is full AND no higher destination → do not move up

$\text{Cost} = \alpha * \text{waiting_time} + \beta * \text{energy} + \gamma * \text{congestion}$

Expected Outcomes: reduced unnecessary trips, better peak handling, and low-latency decisions.

WP2 — Smart Queue: CV + Time-Series

Core Idea: Estimate real-world queue using camera + predict behavior.

- Detect and count people
- Estimate arrival rate
- Predict arrivals and serving rate using LSTM or LightGBM

$q_{\text{next}} = q + \text{arrivals} - \text{served}$

$\text{priority} = \alpha * \text{queue} + \beta * \text{wait} + \gamma * \text{arrival_rate}$

Expected Outcomes: better queue awareness, reduced empty stops, and fair distribution of service.

WP3 — Inclusive HCI / UI / UX

- Icon-based navigation
- Multilingual support
- Voice feedback
- Color-floor mapping
- Accessible controls
- Smart door timing

WP4 — Safety: Multi-Stage AI

Core Principle: Must not miss hazards (high recall), then confirm risk (high precision).

- Layer 1: Hard Safety Rules — overload, obstruction, emergency
- Layer 2: Anomaly Detection — detect abnormal signals early
- Layer 3: Risk Classifier — classify SAFE / CAUTION / UNSAFE
- Layer 4: Decision Gate — high risk → block, medium risk → limit, low risk → continue

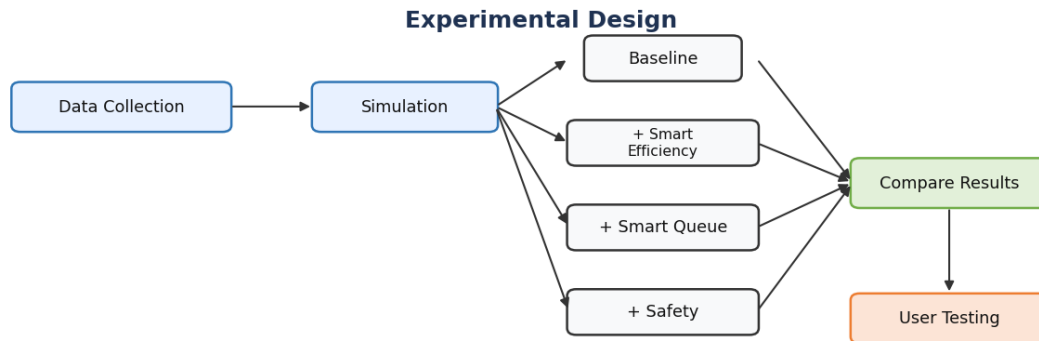
Expected Outcomes: reduced unsafe operation, early detection of faults, and increased system trust.

7. Optimization Objective

Minimize waiting time, energy usage, and safety risk, subject to capacity and safety constraints.

8. Experimental Design

The diagram below follows the original Mermaid-style workflow and is inserted as an image for reliable printing.



9. Expected Contributions

- Hybrid control using rule + ML
- Queue-aware dispatch system
- Safety-aware gating
- Improved UX and accessibility

10. Summary

This proposal introduces a hybrid architecture combining lightweight control, perception, and safety intelligence. By aligning method complexity with problem characteristics, the system improves efficiency, user experience, and operational safety while remaining practical for deployment.